Express Riddler

8 April 2022

Riddle:

In tic-tac-toe, you can rate a square's "real estate value" by how many possible three-in-a-rows pass through it. Thus, the center scores 4, the corners score 3 and the edges score 2.

That means the center is the "most valuable" real estate, while the edges are the "least valuable".

Now, let's try three-dimensional tic-tac-toe, played on a 3-by-3-by-3 board. Which positions are the most and least valuable, and how much are they worth?

Extra credit: What about four-dimensional tic-tac-toe?

Solution:

There are four types of spaces in three-dimensional tic-tac-toe: one center, six face centers, twelve edge centers, and eight vertices, for a total of $3^3 = 27$ spaces. The center space has three orthogonal three-in-a-rows, six along a single diagonal, and four along two diagonals, for a total score of 13. Each face-center space has three orthogonal three-in-a-rows and two along a single diagonal, for a total score of 5. Each edge-center space has three orthogonal three-in-a-rows and one along a single diagonal, for a total score of 4. Each vertex space has three orthogonal three-in-a-rows, three along a single diagonal, and one along two diagonals, for a total score of 7.

There are five types of spaces in four-dimensional tic-tac-toe: one center, eight "cube" centers, 24 face centers, 32 edge centers, and 16 vertices, for a total of $3^4 = 81$ spaces. The center space has four orthogonal three-in-a-rows, 12 along a single diagonal, 16 along two diagonals, and eight along three diagonals, for a total score of 40. Each cube-center space has four orthogonal three-in-a-rows, six along a single diagonal, and four along two diagonals, for a total score of 14. Each face-center space has four orthogonal three-in-a-rows and three along a single diagonal for a total score of 7. Each edge-center space has four orthogonal three-in-a-rows, three along a single diagonal, and one along two diagonals, for a total score of 8. Each vertex space has four orthogonal three-in-a-rows, six along a single diagonal, four along two diagonals, and one along three diagonals, for a total score of 15.